

Computing reachable sets for uncertain nonlinear monotone systems

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Abstract

We address nonlinear reachability computation for uncertain monotone systems, those for which flows preserve a suitable partial orderings on initial conditions. In a previous work [1], we introduced a nonlinear hybridization approach to nonlinear continuous reachability computation. By analysing the signs of off-diagonal elements of system's Jacobian matrix, a hybrid automaton can be obtained, which yields component-wise bounds for the reachable sets. One shortcoming of the method is induced by the need to use whole sets for addressing mode switching. In this paper, we improve this method and show that for the broad class of monotone dynamical systems, component-wise bounds can be obtained for the reachable set in a separate manner. As a consequence, mode switching no longer needs to use whole solution sets. We give examples which show the potentials of the new approach.

Key words: Hybrid systems. Interval analysis. Non-linear systems. Reachability. Uncertain systems. Monotone systems.

1. Introduction

Hybrid systems are complex dynamical systems, which involve the interaction of discrete and continuous dynamics. They are often components of safety-critical systems. It is then necessary to have a thorough and guaranteed insight of the properties of a hybrid system, such as performance, safety or stability [2, 3, 4]. The verification of these properties can be achieved through reachability analyses, some of which require an explicit computation of the hybrid state space, i.e. the set of all trajectories of the hybrid system starting from a possible initial set and under all admissible disturbances and variations in parameter values. A key issue lays then in the calculation of the continuous reachable set for each mode, which boils down to the computation of reachable set for uncertain continuous dynamical systems.

Consider an *uncertain* dynamical system described by non-autonomous differential equations of the following form:

$$\dot{x}(t) = f(x, p, t), \quad x(t_0) \in \mathcal{X}_0 \subseteq \mathcal{D}, \quad p \in \mathcal{P}, \quad t \in \mathcal{I}, \quad (1)$$

where $f : \mathcal{D} \times \mathcal{P} \times \mathbb{R}^+ \mapsto \mathbb{R}^n$ is possibly nonlinear, $\mathcal{D} \subseteq \mathbb{R}^n$, $\mathcal{X}_0 \subseteq \mathbb{R}^m$ is the initial domain for state vector $x(t_0)$ at time $t_0 \geq 0$, $\mathcal{I}$ is the positive interval of existence, and $\mathcal{P}$ is an uncertainty domain for parameter vector p . Moreover, it is assumed that the parameter vector is fixed within $\mathcal{P}$. In this work, because of the methods used, we assume that the uncertainty sets are axis-aligned boxes.

Let us denote by $x(t; t_0, x_0, p)$ or simply $x(t)$ when it is clear from the context, the solution of (1) at time t originating from initial condition x_0 at t_0 and with parameter vector p , and by $\mathcal{X}(t; t_0, \mathcal{X}_0, \mathcal{P})$ or simply $\mathcal{X}(t)$, the set of solutions of (1) at time t originating from each initial condition in $\mathcal{X}_0$ at t_0 and for each parameter vector in $\mathcal{P}$.

The *forward* reachable set of system (1) over a time interval $[t_0, t] \subseteq \mathcal{I}$ is defined as follows

$$\begin{aligned} \mathcal{R}([t_0, t]; \mathcal{X}_0) &\triangleq \{ x(\tau), t_0 \leq \tau \leq t \mid \dot{x}(\tau) = f(x, p, \tau) \wedge x(t_0) \in \mathcal{X}_0 \wedge p \in \mathcal{P} \} \\ &= \bigcup_{\tau \in [t_0, t]} \mathcal{X}(\tau; t_0, \mathcal{X}_0, \mathcal{P}). \end{aligned} \quad (2)$$

Several methods have been developed recently for the explicit computation of reachable sets. When the continuous dynamics are linear, these methods combine time discretization, numerical integration and computational geometry. They use various representations for the reachable sets such as polytopes [2, 5, 6, 7], zonotopes [8] or ellipsoids [9, 10]. Some other methods proceed with hybrid abstractions : the continuous state space is divided into a finite number of cells defined by linear inequalities and then compute reachable set using discrete reachability tools [4, 11, 12, 13]. In [14], a method is given for constructing such hybrid abstractions for polynomial hybrid systems. The case of uncertainty in model parameters has been addressed recently in [15], where a discrete abstraction is used for the analysis of multi-affine uncertain differential equations.

When the continuous dynamics are modelled with a nonlinear differential equation, the computation of the reachable set becomes much harder, which forms one of the main obstacle in safety verification of hybrid systems [12]. A first type of methods rely on a hybridization of the nonlinear continuous-time models, i.e. the use of piecewise simpler, possibly affine or

polynomial conservative approximations of the analysed system on cells defined on the state space [14, 16]. Even though this type of reachability computation methods scale polynomially with the continuous state dimension, they are quite ineffective when used with truly nonlinear systems, mainly because of the over-approximations induced by linearization. A second type of approaches for nonlinear systems compute backward reachable sets using level set methods and viscosity solutions to Hamilton-Jacobi-Isaacs (HJI) partial differential equation [3, 17]. In [18], HJI equations are solved using viability theory. HJI based methods are very effective since they can represent non-convex reachable sets and can naturally handle uncertain time-varying nonlinear dynamics. However they scale exponentially with the continuous state dimension. Hence they are practical only for problems with continuous state variable of small dimension. A third type of approaches for nonlinear systems relies on set integration via interval Taylor methods (ITM) (see the review [19]). ITM have already been used for computing the reachable set of nonlinear continuous dynamical systems in the context of the verification of hybrid systems [20], but no parameter uncertainty were considered. In [21], ITM were used for rigorous simulation of hybrid systems, and an effective technique was also developed to enclose as tightly as possible mode switching points. In [22], ITM were used for simulating dynamical systems with state-dependent switching characteristics, where the dimension of vectors were small. Nevertheless, ITM can be used with uncertain systems in some cases only. Generally, when used with significant uncertainty in either initial state or parameter vectors, the size of the enclosures given by ITM blows up after few integration steps.

In a previous work [1], we introduced a *hybrid bounding* approach to nonlinear reachability computation. Our method relies on the classical Müller's existence theorem [23, 24, 25], i.e. a *comparison theorem* for differential inequalities. Given a time grid $t_0 < t_1 < t_2 < \dots < t_{n_T}$, our method analyzes the signs of the partial derivatives $(\partial f_i / \partial x_j)(x, p, t)$ and $(\partial f_i / \partial p_k)(x, p, t)$, evaluated for any $t \in [t_i, t_{i+1}]$, any $x \in \mathcal{R}([t_i, t_{i+1}])$, and any $p \in \mathcal{P}$. Over each time interval $[t_i, t_{i+1}]$, where these signs remain constant, our method builds two dynamical systems, where there is no uncertainty in either state or parameter vectors, which bracket the actual uncertain system. Hence, the flow pipe of the actual uncertain system is enclosed between a minimal and a maximal solution, obtained as the solution of a system of coupled differential equations. Over each time interval $[t_i, t_{i+1}]$, where the sign of at least one partial derivative changes, our method reverts to classical ITM approaches and computes the reachable set using whole domains. Finally, when the obtained bounding systems are analysed over the whole time interval $[t_0, t_{n_T}]$, they behave as the subsystems of a hybrid system, which switches from one subsystem to another each time a partial derivative changes sign. In fact, the ability of our *hybrid bounding* approach to yield effective results in general, is driven by its ability to ascertain the signs of the partial derivatives. Therefore, when the size of the domains taken for initial state vector or parameter vector are large, one expects the method to have difficulties to ascertain the signs, and hence to seldom use the bounding systems approach. A simple idea to circumvent such a drawback may consist in allowing the domains to be partitioned in a way that renders easier the determination of the signs of the partial derivatives. But then, method complexity would grow exponentially.

The aim of this paper is to show that, for the broad class of monotone dynamical systems, a property to be defined later, we can make mode switching for our *hybrid bounding* approach more efficient. Indeed, for this type of systems, the minimal and maximal solutions can be computed separately, i.e. by solving a system of decoupled differential equations. Consequently, one does not need to compute reachable sets using whole solution sets, neither use the latter when addressing mode switching.

The organisation of the paper is as follows. Sect. 2 overviews ITM for guaranteed set integration and recalls how to use them for computing reachable sets. Sect. 3 introduces monotone dynamical systems, guaranteed set integration for these systems using a comparison theorem and a rule for building bracketing systems for such systems, when the signs of the partial derivatives are constant, then how to compute the reachable sets. Sect. 4 shows how the hybrid bounding approach, introduced in [1], can be improved to deal with monotone dynamical systems. In particular, we show how to use it to compute either maximal or minimal solutions for (1), i.e. how to compute component-wise either upper or lower bounds for the reachable set of an uncertain nonlinear monotone system. Two examples are given in Sect. 5.

2. Computing reachable sets using interval Taylor methods

In this section, we recall how to address guaranteed set integration using interval Taylor methods and how to use it for computing reachable sets.

2.1. Interval analysis

Interval analysis was initially developed to account for the quantification errors introduced by the floating point representation of real numbers with computers and was extended to validated numerics ([26] and the references therein). A real interval $\mathbf{a} = [\underline{a}, \bar{a}]$ is a connected and closed subset of $\mathbb{R}$. We have $\inf(\mathbf{a}) = \underline{a}$ and $\sup(\mathbf{a}) = \bar{a}$. The midpoint of an interval is defined by $\text{mid}(\mathbf{a}) = (\bar{a} + \underline{a})/2$. The set of all real intervals of $\mathbb{R}$ is denoted by $\mathbb{IR}$. Real arithmetic operations are extended to intervals. Consider an operator $\circ \in \{+, -, *, \div\}$ and $\mathbf{a}$ and $\mathbf{b}$ two intervals. Then: $\mathbf{a} \circ \mathbf{b} = [\inf_{u \in \mathbf{a}, v \in \mathbf{b}} u \circ v, \sup_{u \in \mathbf{a}, v \in \mathbf{b}} u \circ v]$.

An interval vector is the Cartesian product of n intervals. The set of n -dimensional real interval vectors is denoted by $\mathbb{IR}^n$. For a n -dimensional real vector $u \in \mathbb{R}^n$ and n -dimensional real interval vectors $\mathbf{a}, \mathbf{b} \in \mathbb{IR}^n$, we have $u \in \mathbf{a} \Leftrightarrow \forall i u_i \in \mathbf{a}_i$ and $\mathbf{a} \subseteq \mathbf{b} \Leftrightarrow \forall i \mathbf{a}_i \subseteq \mathbf{b}_i$.

Consider $g : \mathbb{R}^n \mapsto \mathbb{R}^m$. The range of this function over an interval vector $\mathbf{a}$ is given by $\mathbf{g}(\mathbf{a}) = \{g(u) \mid u \in \mathbf{a}\}$. The interval function $\mathbf{g} : \mathbb{IR}^n \mapsto \mathbb{IR}^m$ is an inclusion function for g if $\forall \mathbf{a} \in \mathbb{IR}^n, \text{range}(\mathbf{g}, \mathbf{a}) \subseteq \mathbf{g}(\mathbf{a})$. An inclusion function for $\mathbf{g}$ can be obtained by replacing each occurrence of a real variable by the corresponding interval and each standard function by its interval counterpart. The resulting function is called the natural inclusion function. The performances of this inclusion function depend on the formal expression for g .

Given a bounded set $\mathcal{E}$ of complex shape, one usually defines an axis-aligned box or a paving, i.e. a union of non-overlapping boxes, $\bar{\mathcal{E}}$ which contains the set $\mathcal{E}$: this is known as an *outer* approximation of it. Likewise, one can also define an *inner* approximation $\underline{\mathcal{E}}$ which is contained in the set $\mathcal{E}$. Hence, we have the following property $\underline{\mathcal{E}} \subseteq \mathcal{E} \subseteq \bar{\mathcal{E}}$.

2.2. Interval Taylor methods

Consider now the differential equation (1), where $\mathcal{X}_0 = \mathbf{x}_0 = [\underline{x}_0, \bar{x}_0]$ and $\mathcal{P} = \mathbf{p} = [\underline{p}, \bar{p}]$. Define a time grid $t_0 < t_1 < t_2 < \dots < t_{n_T}$ which is not necessarily equally spaced. The objective is to compute interval vectors $\mathbf{x}_j, j = 1, \dots, n_T$, that are *guaranteed* to contain the set of solutions $\mathcal{X}(t_j)$ of (1) at time t_j , originating from each initial condition in $\mathbf{x}_0$ at t_0 and for each parameter vector in $\mathbf{p}$.

Definition 1 (A priori enclosure). A vector interval $\tilde{\mathbf{x}}_j$ which satisfies the property

$$\forall t \in [t_j, t_{j+1}], \mathcal{X}(t) \subset \tilde{\mathbf{x}}_j \quad (3)$$

is an a priori enclosure for the solutions of (1) over the time interval $[t_j, t_{j+1}]$. $\square$

The computation of the set of solutions $\mathcal{X}(t_j)$ can be solved efficiently by using interval Taylor expansions. These interval methods are usually one-step methods which proceed in two phases.

1. They first verify existence and uniqueness of the solution using the Banach fixed point theorem and the Picard-Lindelöf operator. Though this theoretical justification is done in the literature for non parametric ODEs, they still hold in presence of an additional parameter vector, since we can define a new non parametric ODE using the extended state vector $z = (x, p)$ and the extended system $\dot{z} = h(z) \equiv (f(x, p, t), 0)$. Interval Taylor methods compute an a priori enclosure $\tilde{\mathbf{x}}_j$ which satisfies (3) and adapt integration time step size $\Delta t_j = t_{j+1} - t_j$ if necessary. A simple method for obtaining the a priori enclosure uses $\tilde{\mathbf{x}}_j = \mathbf{x}_j + [0, \Delta t_j]f(\mathbf{x}_j, \mathbf{p}, [t_j, t_{j+1}])$ as initial guess, then the set $\tilde{\mathbf{x}}_j$ is widened and/or the time step Δt_j is reduced until the following inclusion is satisfied [27]

$$\mathbf{x}_j + [0, \Delta t_j]f(\tilde{\mathbf{x}}_j, \mathbf{p}, [t_j, t_{j+1}]) \subseteq \tilde{\mathbf{x}}_j. \quad (4)$$

There are more efficient techniques, which are detailed in [19, 28].

2. Then, they compute a tighter enclosure $\mathbf{x}_{j+1}$ of the set of solutions of (1) at t_{j+1} , i.e.

$$\mathcal{X}(t_{j+1}) \subseteq \mathbf{x}_{j+1} = \mathbf{x}_j + \sum_{i=1}^{k-1} \Delta t_j^i f^{[i]}(\mathbf{x}_j, \mathbf{p}, t_j) + \Delta t_j^k f^{[k]}(\tilde{\mathbf{x}}_j, \mathbf{p}, [t_j, t_{j+1}]) \quad (5)$$

which corresponds to a Taylor expansion of order k where $\tilde{\mathbf{x}}_j$ is used to compute the remainder term. The coefficients $f^{[i]}$ are the Taylor coefficients of the solution $x(t)$ which are usually computed via automatic differentiation.

The enclosures thus obtained are said *validated*, which is in contrast to conventional numerical integration techniques, which derive approximations with unknown global error and where the accumulation of both truncation and round-off errors may cause the computed solution to deviate widely from the real one. It is well-known that the explicit scheme (5) is width-increasing and thus not suitable for numerical implementation. To solve such a drawback, one can use mean-value forms, matrix preconditioning, and linear transformations

(see the review in [19]). For instance, a popular method by Lohner uses at each time step a point vector $\hat{x}_j$, a point matrix A_j and an interval vector $\mathbf{r}_j$ such that $\mathbf{x}_j = \{\hat{x}_j + A_j \mathbf{r}_j \mid \mathbf{r}_j \in \mathbf{r}_j\}$. Then it tunes the algorithm such that over-approximations impact only $\mathbf{r}_j$. In this method, matrices A_j are obtained via QR factorization [29]. In [30, 31], a Taylor series expansion with respect to initial state has also been used. A more general scheme has been developed in [32], where the interval method is founded on the Hermite-Obreshkoff expansion series, where the sought enclosure appears both implicitly and explicitly. In [33], an alternative technique has been introduced, where constraint propagation techniques are used in connection with a guaranteed relaxation of the ODE in order to build a pruning step. Finally, there are open source softwares available that implement most of the above techniques (e.g. Nedialkov's VNODE-LP, Rauh and Auer's VALENCIA-IVP and Makino and Berz's COSY INFINITY).

2.3. Over-approximating reachable sets

By using interval Taylor methods one can obtain guaranteed enclosures for the set of solutions of (1) at grid points t_j , $j = 0, \dots, n_T$. It remains to enclose the set of solutions between two grid points. We will show now how to derive explicit formulas which characterize the boundaries of the reachable set for any time.

For $j = 0, \dots, n_T - 1$ and $t \in [t_j, t_{j+1}]$ define

$$\mathbf{x}(t) = \mathbf{x}_j + \sum_{i=1}^{k-1} (t - t_j)^i f^{[i]}(\mathbf{x}_j, \mathbf{p}, t_j) + (t - t_j)^k f^{[k]}(\boldsymbol{\xi}_j, \mathbf{p}, [t_j, t_{j+1}]) \quad (6)$$

Proposition 1.

$$\text{If } \boldsymbol{\xi}_j \supseteq \tilde{\mathbf{x}}_j \Rightarrow \forall t \in [t_j, t_{j+1}], \mathcal{X}(t) \subseteq \mathbf{x}(t), \quad j = 0, \dots, n_T - 1 \quad (7)$$

Proof 1. It suffices to write a Taylor series expansion at time t_j and use $\tilde{\mathbf{x}}_j$ for evaluating the remainder term (see [19]).

Define $\overline{\mathcal{R}}$ as an over-approximation of a reachable set $\mathcal{R}$, as follows

$$\forall t, t' \in [t_0, t_{n_T}], \overline{\mathcal{R}}([t, t']; \mathbf{x}(t)) \supseteq \mathcal{R}([t, t']; \mathbf{x}(t)) \quad (8)$$

Proposition 2.

$$\forall t \in [t_j, t_{j+1}], \overline{\mathcal{R}}([t_j, t]; \mathbf{x}_j) = \cup_{\tau \in [t_j, t]} \mathbf{x}(\tau), \quad j = 0, \dots, n_T - 1 \quad (9)$$

and satisfies

$$\forall t \in [t_j, t_{j+1}], \overline{\mathcal{R}}([t_j, t]; \mathbf{x}_j) \subseteq \tilde{\mathbf{x}}_j, \quad j = 0, \dots, n_T - 1 \quad (10)$$

Proof 2. Obvious from (7).

Define $\overline{\mathcal{R}}([t_0, t_0]; \mathbf{x}_0) = \mathbf{x}_0$.

Proposition 3. *An over-approximation of the reachable set (2) is given by*

$$\forall t \in [t_j, t_{j+1}], \overline{\mathcal{R}}([t_0, t]; \mathbf{x}_0) = \overline{\mathcal{R}}([t_0, t_j]; \mathbf{x}_0) \cup \overline{\mathcal{R}}([t_j, t]; \mathbf{x}_j), \quad j = 1, \dots, n_T - 1 \quad (11)$$

and satisfies

$$\forall t \in [t_j, t_{j+1}], \overline{\mathcal{R}}([t_0, t]; \mathbf{x}_0) \subseteq \cup_{i \in \{0, j\}} \tilde{\mathbf{x}}_i, \quad j = 1, \dots, n_T - 1 \quad (12)$$

Proof 3. *Obvious from (7) and proposition (2).*

As a conclusion, it is clear that due to (6) and (11), one can derive explicit formulas that characterize the boundaries of the reachable set. In practice however, one can use instead of (11) the over-approximation (12) obtained by using the *a priori* solutions $\tilde{\mathbf{x}}_j$ only.

Remark 1. When the size of the initial domain for state vector or the parameter vector box is large, guaranteed numerical integration is often doomed to diverge. In such cases, pessimism might be controlled by bisection, i.e. performing a partition of the initial state vector or parameter vector domains. Nevertheless, such a procedure scales exponentially with the dimension of the state vector and hence increases computation times significantly. The method introduced in this paper investigates the possibility to solve this problem when the system under study is a monotone dynamical system, without using bisection. $\square$

3. Computing reachable sets for monotone systems

In this section, we overview monotone systems. We show how to use comparison principles for monotone systems for set integration, and then for computing reachable sets.

3.1. Monotone systems

Monotone dynamical systems are those for which flows preserve a suitable partial ordering on states, hence on *initial conditions*. This is a broad class of dynamical systems, which is met in many chemical, physical and economic models, and is one of the most important classes of systems in the theoretical biology [34].

The theory of monotone systems is well known and has been mainly developed by Hirsch and Smith after the seminal work of Müller, Kamke and Krasnoselskij (see [35, 36, 37] and the references therein). It has been extended to systems with inputs and outputs by Angeli and Sontag [34, 38].

Definition 2 (monotone systems [36, 37, 39, 34]). *Let us assume that system (1) is forward-complete (solutions do not blow up in finite time). Given a partial ordering on $\mathcal{D}$ and $p \in \mathcal{P}$, system (1) is monotone with respect to this partial ordering, if for all initial state vectors x_1, x_2 , the following property holds:*

$$\text{if } x_1 \preceq x_2 \text{ then } \forall t \geq t_0, x(t; t_0, x_1, p) \preceq x(t; t_0, x_2, p) \quad (13)$$

$\square$

The partial ordering considered is the one induced by closed proper cones $K \subseteq \mathbb{R}^n$ in the sense that $x \preceq y$ iff $y - x \in K$. The cones K are assumed to have a nonempty interior and are pointed.

We will consider in the sequel the particular case when K is an orthant cone in $\mathbb{R}^n$, i.e., a set S_ϵ of the form $\{x \in \mathbb{R}^n \mid \epsilon_i x_i \geq 0\}$, where $\epsilon_i = \pm 1$ for each i . A useful test for monotonicity with respect to arbitrary orthant cones is given in [39]. This test uses a graph theoretical approach as follows.

Assume that all off-diagonal elements of the Jacobian matrix, i.e. the partial derivatives $(\partial f_i / \partial x_j)(x, p, t)$ for $i \neq j$ do not change sign, i.e. they are always ≥ 0 or always ≤ 0 . Then we can associate to the system a signed directed *incidence graph* G with a vertex for each component, and with vertices connected by signed, directed edges labelled “+” or “−”, whose labels are determined by the signs of the appropriate Jacobian matrix entry (ignoring diagonal elements of the Jacobian matrix). The edges between vertices are built as follows.

1. If $(\partial f_i / \partial x_j)(x, p, t) > 0$ for all $x \times p \times t \in \mathcal{D} \times \mathcal{P} \times \mathcal{I}$, then we draw a positive edge e_{ij}^x directed from vertex x_j to x_i .
2. If $(\partial f_i / \partial x_j)(x, p, t) < 0$ for all $x \times p \times t \in \mathcal{D} \times \mathcal{P} \times \mathcal{I}$, then we draw a negative edge e_{ij}^x directed from vertex x_j to x_i .
3. Otherwise, $(\partial f_i / \partial x_j)(x, p, t) = 0$ for all $x \times p \times t \in \mathcal{D} \times \mathcal{P} \times \mathcal{I}$, and no edge from x_j to x_i is drawn.

A *cycle*, not necessarily directed, is a finite sequence of vertices $v_{n_0}, v_{n_1}, \dots, v_{n_L}$, such that $v_{n_L} = v_{n_0}$ and the constraints that either e_{ij}^v or e_{ji}^v is an edge whenever v_i and v_j appear consequently in the cycle. The *sign of a cycle* is defined as the product of the signs of the edges comprising it.

Theorem 1 (Graphical condition for monotonicity [39]). *A system (1) which admits an incidence graph G built according to the above set of rules is monotone with respect to some orthant cone in $\mathcal{D}$ if and only if its incidence graph does not contain any negative cycle.* $\square$

Now, if the cone is the main orthant ($\epsilon = (1, \dots, 1)$), the requirement is that all partial derivatives must be nonnegative with possible exception for the diagonal terms of the Jacobian of f with respect to x . A monotone system with respect to the main orthant is called a *cooperative system* [36].

Property 1 (Converting a monotone system to a cooperative one). *It is interesting to note that when the orthants are defined in the form S_ϵ , i.e., by restricting the sign of each coordinate, a monotone system with respect to any orthant S_ϵ can be converted into a cooperative system in the main orthant, by the coordinate change $z = \text{Diag}(\epsilon)x$, where $\text{Diag}(\epsilon)$ is a diagonal matrix of diagonal elements ϵ_i [37].* $\square$

Property 2 (Comparison principle for cooperative systems). *Consider two cooperative systems*

$$\dot{z}_1(t) = f_1(z_1, p, t), \quad z_1(t_0) = z_{1,0} \quad (14)$$

$$\dot{z}_2(t) = f_2(z_2, p, t), \quad z_2(t_0) = z_{2,0}, \quad (15)$$

where z_1 and $z_2 \in \mathcal{D} \subseteq \mathbb{R}^n$, $p \in \mathcal{P}$. Then the following property holds:

$$\begin{aligned} \text{if } (z_{1,0} \leq z_{2,0}) \wedge (\forall z, \forall p, f_1(z, p, t) \leq f_2(z, p, t)) \\ \Rightarrow \forall t \geq t_0, z_1(t; t_0, z_{1,0}, p) \leq z_2(t; t_0, z_{2,0}, p) \end{aligned} \quad (16)$$

□

3.2. Guaranteed set integration with cooperative systems

Given a dynamical system which is monotone with respect to any orthant S_ϵ , we can use the coordinate change suggested in property 1, in order to convert it to a cooperative system. Then, we can use property 2 in order to proceed with set integration.

Theorem 2 (Guaranteed set integration using cooperative systems [36, 40]). *If there exists a pair of cooperative systems $\dot{\bar{x}} = \bar{f}(\bar{x}, \bar{\pi}, t)$ and $\dot{\underline{x}} = \underline{f}(\underline{x}, \underline{\pi}, t)$ which satisfy*

$$\forall p \in \mathcal{P}, \forall x \in \mathcal{D}, \forall t \geq t_0, \underline{f}(\underline{x}, \underline{\pi}, t) \leq f(x, p, t) \leq \bar{f}(\bar{x}, \bar{\pi}, t), \quad (17)$$

where $\bar{\pi}$ and $\underline{\pi}$ are parameter vectors belonging to $\mathcal{P}$, states $\bar{x}(t)$ and $\underline{x}(t)$ evolve in $\mathcal{D}$. Moreover, if these cooperative systems can be given initial conditions $\bar{x}(\bar{\pi}, t_0)$ and $\underline{x}(\underline{\pi}, t_0)$ such that

$$\forall p \in \mathcal{P}, \quad \underline{x}(\underline{\pi}, t_0) \leq x(t_0) \leq \bar{x}(\bar{\pi}, t_0), \quad (18)$$

then the solution of (1) satisfies

$$\forall t \geq t_0, \mathcal{X}(t) \subseteq [\underline{x}(t), \bar{x}(t)], \quad (19)$$

where $\bar{x}(t)$ is the solution of

$$\dot{\bar{x}} = \bar{f}(\bar{x}, \bar{\pi}, t), \quad \bar{x}(t_0) = \bar{x}(\bar{\pi}, t_0), \quad (20)$$

and $\underline{x}(t)$ is the solution of

$$\dot{\underline{x}} = \underline{f}(\underline{x}, \underline{\pi}, t), \quad \underline{x}(t_0) = \underline{x}(\underline{\pi}, t_0) \quad (21)$$

□

Now, the bracketing systems (20–21) involve non uncertain quantities. Therefore interval Taylor methods can be used for guaranteed numerical evaluation of $\underline{x}(t)$ and $\bar{x}(t)$. Indeed, the overestimations in interval computations can be efficiently controlled by the methods of section 2.2.

The main difficulty is to obtain suitable bracketing systems in the general case. However, when system (1) is cooperative (after a possible change of coordinates according to property 1), we can extend the rules introduced in [1, 41, 42].

Rule 1 (A rule for bracketing cooperative systems). Assume that function f is differentiable w.r.t. p . Denote $\mathcal{P}$ as the interval vector $\mathbf{p}$. Define the scalar function $\bar{\delta}^i(p_k)$ as follows

$$\begin{aligned} \text{if } (\forall p \in \mathbf{p}, \forall x \in \mathcal{X}(t), \forall t \geq t_0, (\partial f_i / \partial p_k)(x, p, t) \geq 0), \quad \text{then } \bar{\delta}^i(p_k) = \bar{p}_k, \\ \text{else } \bar{\delta}^i(p_k) = \underline{p}_k. \end{aligned} \quad (22)$$

In a similar way, define the scalar function $\underline{\delta}^i(p_k)$ as follows

$$\begin{aligned} \text{if } (\forall p \in \mathbf{p}, \forall x \in \mathcal{X}(t), \forall t \geq t_0, (\partial f_i / \partial p_k)(x, p, t) \geq 0), \quad \text{then } \underline{\delta}^i(p_k) = \underline{p}_k, \\ \text{else } \underline{\delta}^i(p_k) = \bar{p}_k. \end{aligned} \quad (23)$$

Furthermore, define the vectors

$$\bar{\Delta}^i(p) = [\bar{\delta}^i(p_1), \dots, \bar{\delta}^i(p_k), \dots]^T \quad (24)$$

and

$$\underline{\Delta}^i(p) = [\underline{\delta}^i(p_1), \dots, \underline{\delta}^i(p_k), \dots]^T. \quad (25)$$

If system (1) is cooperative over $\mathcal{D}$, then the upper bracketing system (20) can be obtained as follows

$$\dot{\bar{x}}_i(t) = f_i(\bar{x}, \bar{\Delta}^i(p), t), \quad \bar{x}_i(t_0) = \bar{x}_{0,i}, \quad i = 1, \dots, n \quad (26)$$

and the lower bracketing system (21) as follows

$$\dot{\underline{x}}_i(t) = f_i(\underline{x}, \underline{\Delta}^i(p), t), \quad \underline{x}_i(t_0) = \underline{x}_{0,i}, \quad i = 1, \dots, n. \quad (27)$$

Note that the bracketing systems (26-27) are decoupled dynamical systems. That is, maximal solution $\bar{x}(t)$ and minimal solution $\underline{x}(t)$ can be obtained separately. Now, to derive validated results, we can solve both differential equations using interval Taylor methods described in section 2.2.

3.3. How tight are the enclosures

An important issue when deriving enclosures for the solution of (1) as suggested in theorem 2 and by using rule 1 is how tight are the enclosures given by (26-27).

The derived enclosures for the reachable set (2) are *tight*, if they give interval state vectors at each time t , which can be actually reached by system (1) with the given initial and parameter intervals. Obviously, this is true if systems (26) and (27) are feasible, which in turn is true if there exists p_1 and p_2 in $[\underline{p}, \bar{p}]$ such that $\bar{x}(t)$ is the solution of

$$\dot{\bar{x}}(t) = f(\bar{x}, p_1, t), \quad \bar{x}(t_0) = \bar{x}_0, \quad (28)$$

and $\underline{x}(t)$, the solution of

$$\dot{\underline{x}}(t) = f(\underline{x}, p_2, t), \quad \underline{x}(t_0) = \underline{x}_0. \quad (29)$$

If a parameter vector component p_i appears in the algebraic expression of several functions f_j , then, it is likely that the upper system $\bar{f}$ is built using $\bar{p}_i$ for f_j and using $\underline{p}_i$ for f_k . In this case, upper system (26) is not feasible. This will also be the case for lower system (27). Consequently, the derived enclosures will be *not* tight.

Nevertheless, when the system under study is cooperative (possibly after a change of coordinate) and the bracketing systems built using (26) and (27), then the size of the derived enclosures are usually smaller than the ones obtained via ITM. Indeed, when one uses ITM, the uncertain dynamics thus exhibited by interval inclusion functions, captures the dynamics of the *certain* (with no uncertainty) bracketing systems. Consequently, the results of the latter are always included in ITM results.

3.4. The reachable sets

Using ITM for solving systems (26) and (27), one obtains an enclosure for the solution of (1) as follows

$$\forall t \in [t_0, t_{n_T}], \quad \mathbf{x}(t) = [\text{Inf}(\underline{\mathbf{x}}(t)), \text{Sup}(\bar{\mathbf{x}}(t))]. \quad (30)$$

In addition, it is easy to prove that the a priori enclosures $\tilde{\mathbf{x}}_j$ for $\mathbf{x}(t)$, $t \in [t_j, t_{j+1}]$ as defined by (3) can also be obtained as follows

$$\tilde{\mathbf{x}}_j = [\text{Inf}(\tilde{\underline{\mathbf{x}}}_j), \text{Sup}(\tilde{\bar{\mathbf{x}}}_j)], \quad j = 1, \dots, n_T - 1. \quad (31)$$

where $\tilde{\underline{\mathbf{x}}}_j$ and $\tilde{\bar{\mathbf{x}}}_j$ are a priori enclosures obtained for $\underline{\mathbf{x}}(t)$ and $\bar{\mathbf{x}}(t)$. Finally, one can use the enclosures $\mathbf{x}(t)$ obtained via (30) in (6) or (11) and (31) in (12) in order to compute an over-approximation for the reachable set.

The main difficulty is now to obtain suitable bracketing functions (26)-(27) in the general case. That is, when the signs of the partial derivatives change over the time period under study.

4. Computing reachable sets for cooperative systems using hybrid automata as bracketing systems

In this section, we address the case of uncertain monotone dynamical systems for which the sign of the partial derivatives $(\partial f_i / \partial p_k)(x, p, t)$ may vary over the time period under study. We have addressed this issue in the general case in [1] : a hybrid bounding approach was defined which uses rule 1 over each time interval $[t_i, t_{i+1}]$, where these signs remain constant and revert to ITM over each time interval $[t_i, t_{i+1}]$, where the sign of at least one partial derivative changes. In the latter case, the computation of the reachable set is done using whole domains. Finally, when the obtained systems are analysed over the whole time interval $[t_0, t_{n_T}]$, they behave as the subsystems of a hybrid system, which switches from one subsystem to another each time a partial derivative changes sign.

When the system under study is cooperative, (or monotone and after a proper change of coordinates), we have shown in subsection 3.2 that the upper and lower component-wise bounds of the reachable set can be computed separately. This is an interesting property

since mode switching in the hybrid bounding approach needs not be done with the whole reachable set but only with its component-wise bounds.

In the sequel, we address the computation of component-wise bounds for the reachable set of cooperative systems using an adapted version of our hybrid bounding approach. We will focus on the upper bound, the lower one being obtained in a similar manner. Furthermore, we will also improve the rule for building the bracketing systems.

4.1. A hybrid automaton as upper bracketing system

The issue is to obtain and solve system (20), i.e. component-wise upper bounds for the upper bracketing system

$$\dot{\bar{x}} = \bar{f}(\bar{x}, p, t), \quad \bar{x}(t_0) = \bar{x}_0, \quad p \in \mathbf{p}, \quad t \in \mathcal{I} \quad (32)$$

Denote by $\mathcal{H}$ the hybrid system characterizing the upper bracketing system for (32). Denote by $\mathcal{Q} \subset \mathbb{N}$ the finite set of modes of $\mathcal{H}$ over which one subsystem is active, and by $\mathcal{E} \subseteq \mathcal{Q} \times \mathcal{Q}$ the set of mode transitions. Denote by t_e a switching time instant and by $\mathcal{S}$ the collection of switching time instants

$$\mathcal{S} = \{ t_e \in [t_0, t_{n_T}] \mid \exists i, \exists r, \exists p \in \mathbf{p}, \exists \bar{x} \in \bar{\mathbf{x}}(t_e) \mid (g_{i,r}(\bar{x}, p, t_e) = 0) \}. \quad (33)$$

where $g_{i,r}(\cdot) = (\partial f_i / \partial p_r)(\cdot)$ with $i \in \{i = 1, \dots, n\}$, $r \in \{i = 1, \dots, m\}$.

The switching time instants are not known a priori, and they depend on initial conditions and parameter vector uncertainty set. This constitutes an issue for this bounding approach method. Along the lines of our previous work [1], we will show now how to solve this problem and how to build the automaton that governs the dynamics of the upper bounding systems.

Let us split the experiment time period $[t_0, t_{n_T}]$ into a succession of integration time intervals $[t_j, t_{j+1}]$, where t_j are the time grid points defined in the preceding sections.

Denote $\mathcal{I}_M$, the set of time intervals $[t_j, t_{j+1}]$ over which no switching occurs, i.e.

$$\mathcal{I}_M = \{[t_j, t_{j+1}] \subseteq [t_0, t_{n_T}] \mid \forall e \in \mathcal{E}, t_e \notin [t_j, t_{j+1}]\}. \quad (34)$$

Similarly, define the set $\mathcal{I}_S$ of intervals where a switching occurs, i.e.,

$$\mathcal{I}_S = \{[t_j, t_{j+1}] \subseteq [t_0, t_{n_T}] \mid \exists e \in \mathcal{E}, t_e \in [t_j, t_{j+1}]\}. \quad (35)$$

Denote by $\underline{\mathcal{I}}_M \subseteq \mathcal{I}_M$, an inner approximation for $\mathcal{I}_M$, and by $\overline{\mathcal{I}}_S \supseteq \mathcal{I}_S$, an outer approximation for $\mathcal{I}_S$. We have

$$[t_0, t_{n_T}] = \mathcal{I}_M \cup \mathcal{I}_S \Rightarrow \overline{\mathcal{I}}_S = [t_0, t_{n_T}] \setminus \underline{\mathcal{I}}_M. \quad (36)$$

The next proposition shows how to obtain $\underline{\mathcal{I}}_M$ and hence $\overline{\mathcal{I}}_S$.

Proposition 4 (Inner approximation of $\mathcal{I}_M$). *An inner approximation $\underline{\mathcal{I}}_M$ is given by*

$$\underline{\mathcal{I}}_M \triangleq \{ [t_j, t_{j+1}] \subseteq [t_0, t_{n_T}] \mid \forall i, \forall k, (0 \notin \mathbf{g}_{i,k}(\widetilde{\mathbf{x}}_j, \mathbf{p}, [t_j, t_{j+1}])) \}, \quad (37)$$

where $\mathbf{g}_{i,k}(\cdot)$ is an inclusion function for $g_{i,k}(\cdot)$.

Proof 4. Since the a priori solution $\tilde{\mathbf{x}}_j$ of (32) as given by (4) encloses the whole state trajectory over $[t_j, t_{j+1}]$, we can write

$$\begin{aligned} \forall t \in [t_j, t_{j+1}], \forall x(t) \in \mathcal{R}([t_j, t_{j+1}]; \tilde{\mathbf{x}}_j), \forall p \in \mathbf{p}, \\ \forall i, \forall k, g_{i,k}(x, p, t) \in \mathbf{g}_{i,k}(\tilde{\mathbf{x}}_j, \mathbf{p}, [t_j, t_{j+1}]). \end{aligned} \quad (38)$$

Consequently

$$\begin{aligned} 0 \notin \mathbf{g}_{i,k}(\tilde{\mathbf{x}}_j, \mathbf{p}, [t_j, t_{j+1}]) \\ \Rightarrow \forall t \in [t_j, t_{j+1}], \forall \bar{x}(t) \in \mathcal{R}([t_j, t_{j+1}]; \tilde{\mathbf{x}}_j), \forall p \in \mathbf{p}, g_{i,k}(\bar{x}, p, t) \neq 0 \end{aligned} \quad (39)$$

This ends the proof.

Now, according to proposition 4, we can use rule 1 over each time interval $\mathbf{I}_m \in \underline{\mathcal{I}}_M$ to obtain the upper bracketing system $\bar{f}_{(m)}$ for (1). Denote by $\tilde{\mathbf{x}}_{(m)}$ the a priori enclosure of the solution $\bar{\mathbf{x}}(t)$ of (32) over the time interval $\mathbf{I}_m$, we have

$$\begin{aligned} \forall \mathbf{I}_m \in \underline{\mathcal{I}}_M, \forall m \in \mathcal{Q}, \forall p \in \mathbf{p}, \forall \bar{x} \in \tilde{\mathbf{x}}_{(m)}, \forall t \in \mathbf{I}_m, \\ f_i(\bar{x}, p, t) \leq \bar{f}_{i,(m)}(\bar{x}, \bar{\Delta}^i(p), t), \quad i = 1, \dots, n. \end{aligned} \quad (40)$$

It remains to deal with time intervals $\mathbf{I}_s \in \overline{\mathcal{I}}_S$, i.e. time intervals which contain $t_e \in \mathcal{S}$ as defined by (33). Since rule 1, and hence the bounding method, cannot be used with system (32), we suggested in [1] to use instead an ITM applied to the uncertain system (32). Here, we will improve the bounding approach by an improved bracketing rule, as follows.

Rule 2 (A modified rule for upper bracketing over $\overline{\mathcal{I}}_S$). Let us consider the same notations and assumptions as in rule 1. Denote the set of indices

$$\mathcal{I}^* = \{i^* \mid \exists k^*, \exists t_e \in \mathbf{I}_S, \exists \bar{x} \in \bar{\mathbf{x}}(t_e), \exists p \in \mathbf{p} \mid g_{i^*,k^*}(\bar{x}, p, t_e) = 0\}, \quad (41)$$

$$\mathcal{K}^*(i^*) = \{k^* \mid \exists t_e \in \mathbf{I}_S, \exists \bar{x} \in \bar{\mathbf{x}}(t_e), \exists p \in \mathbf{p} \mid g_{i^*,k^*}(\bar{x}, p, t_e) = 0\}. \quad (42)$$

Define a modified vector $\bar{\Delta}^{i^*}(p)$ as follows

$$\begin{aligned} \text{if } i \notin \mathcal{I}^* \text{ then } \bar{\Delta}^{i^*}(p) = [\bar{\delta}^i(p_1), \dots, \bar{\delta}^i(p_k), \dots]^T \\ \text{else } \bar{\Delta}^{i^*}(p) = [\bar{\delta}^i(p_1), \dots, \bar{\delta}^i(p_{k^*-1}), p_{k^*}, \bar{\delta}^i(p_{k^*+1}), \dots]^T, \quad k^* \in \mathcal{K}^*(i^*) \end{aligned} \quad (43)$$

For $t \in \mathbf{I}_s$, we can write

$$\forall t \in \mathbf{I}_s, \forall \bar{x} \in \tilde{\mathbf{x}}_s, \forall p \in \mathbf{p}, f_i(\bar{x}, p, t) \leq f_i(\bar{x}, \bar{\Delta}^{i^*}(p), t). \quad (44)$$

For a given $p \in \mathbf{p}$, an upper bracketing system can be obtained for (32) as follows

$$\dot{\bar{x}}_i(t) = f_i(\bar{x}, \bar{\Delta}^{i^*}(p), t), \quad \bar{x}_i(t_0) = \bar{x}_{0,i}, \quad i = 1, \dots, n. \quad (45)$$

Using the modified rule 2, we can now deal with time intervals $\mathbf{I}_s$ and write

$$\forall \mathbf{I}_s \in \overline{\mathcal{I}_S}, \forall s \in \mathcal{Q}, \forall \bar{\mathbf{x}} \in \widetilde{\mathbf{x}}_{(s)}, \forall \mathbf{p} \in \mathbf{p}, \forall t \in \mathbf{I}_s, \\ f_i(\bar{\mathbf{x}}, \mathbf{p}, t) \leq \bar{f}_{i,(s)}(\bar{\mathbf{x}}, \bar{\Delta}^{i*}(\mathbf{p}), t), \quad i = 1, \dots, n. \quad (46)$$

For $i^* \in \mathcal{I}^*$, denote $\bar{\Delta}^{i*}(\mathbf{p}) = [\bar{\delta}^i(p_1), \dots, \bar{\delta}^i(p_{k^*-1}), \mathbf{p}_{k^*}, \bar{\delta}^i(p_{k^*+1}), \dots]^T$, for $k^* \in \mathcal{K}^*(i^*)$. The maximal solution of (32) for $t \in \mathbf{I}_s$, is given by the following uncertain system

$$\forall t \in \mathbf{I}_s, \dot{\bar{\mathbf{x}}}_i = \bar{f}_{i,(s)}(\bar{\mathbf{x}}, \pi_i, t), \quad \bar{\mathbf{x}}(t_j) = \bar{\mathbf{x}}_j, \quad \pi_i \in \bar{\Delta}^{i*}(\mathbf{p}), \quad i = 1, \dots, n. \quad (47)$$

To compute the maximal solution for $t \in \mathbf{I}_s$, it suffices to use ITM with system (47), where the k^* -th component p_k^* is taken uncertain, i.e. $p_k^* \in \mathbf{p}_k$ for all $k^* \in \mathcal{K}^*(i^*)$ in the i^* -th function f_{i^*} , for all $i^* \in \mathcal{I}^*$.

Remark 2. Note that since the widths of the intervals $\mathbf{I}_s$ are equal to an integration time step, the overestimation will be very small in the validated integration method. Eventually, the time intervals $\mathbf{I}_s$ might also be reduced using for instance the methods developed in [21, 43, 44].

Let us denote in the sequel *m-type modes*, the ones which are active over $\mathbf{I}_m$ time intervals, and denote *s-type modes*, the ones which are active over $\mathbf{I}_s$ time intervals.

Proposition 5. Mode switching cannot occur from an *m-type mode* to another *m-type mode*.

Proof 5. This is induced by the rule used for deriving the upper bracketing system. If current mode is of *m-type*, then the sign of all the partial derivatives can be ascertained. If one sign is to change over the current integration time interval, then the next mode is necessarily of *s-type*.

The following propositions make it possible to detect *on-the-fly* the switching between *m-type* and *s-type* modes, and possibly between *s-types*, and to instantiate a new mode.

Proposition 6 (Switching *m-type* $\mapsto$ *s-type*).

$$\text{If } ((m\text{-type mode}) \wedge (\exists i, \exists k, 0 \in \mathbf{g}_{i,k}(\widetilde{\mathbf{x}}_j, \mathbf{p}, [t_i, t_{i+1}])) \\ \Rightarrow (e = (m\text{-type mode}, s\text{-type mode}) \wedge (\widetilde{\mathbf{x}}_j \text{ must be re-computed}))) \quad (48)$$

Proof 6. When active mode is a *m-type mode*, and the inclusion functions in (48) contains 0, then a transition occurs, and the new mode is necessarily a *s-type* one. Indeed in this case, the sign of the partial derivative cannot be ascertained for all t in $[t_j, t_{j+1}]$, rule 2 must be used and ITM applied to the uncertain system (47). Now, recall that $\widetilde{\mathbf{x}}_j$ is computed via (40). But, since solutions $\bar{\mathbf{x}}(t)$ computed with the upper bounding systems derived for current mode is valid only over $[t_j, t_e]$, $\widetilde{\mathbf{x}}_j$ must be re-computed with the uncertain systems (47).

Proposition 7 (Switching s -type $\mapsto m$ -type).

$$\begin{aligned} & \text{If } ((s\text{-type mode}) \wedge (\forall i, \forall k, 0 \notin \mathbf{g}_{i,k}(\tilde{\mathbf{x}}_j, \mathbf{p}, [t_i, t_{i+1}])))) \\ & \Rightarrow (e = (s\text{-type mode}, m\text{-type mode}) \wedge (\tilde{\mathbf{x}}_j \text{ needs not be re-computed})). \end{aligned} \quad (49)$$

Proof 7. When active mode is an s -type mode, and it becomes possible to ascertain the signs of all the partial derivatives in (49) then a transition occurs, and the new mode is necessarily a m -type one. The sign analysis is done using the inclusion functions. $\tilde{\mathbf{x}}_j$ is computed via (40) using ITM. It is always valid. Numerical integration can be taken forward from t_{j+1} .

Proposition 8 (Switching s -type $\mapsto s$ -type). Current mode is of s -type. Denote by $i^* \in \mathcal{I}^*$ and $k^* \in \mathcal{K}^*(i^*)$ the indices such that $0 \in \mathbf{g}_{i^*,k^*}(\tilde{\mathbf{x}}_{j-1}, \mathbf{p}, [t_i, t_{i+1}])$.

$$\begin{aligned} & \text{If } ((s\text{-type mode}) \wedge (\exists i, \exists k, ((i \neq i^*) \vee (k \neq k^*)), 0 \in \mathbf{g}_{i,k}(\tilde{\mathbf{x}}_j, \mathbf{p}, [t_i, t_{i+1}])))) \\ & \Rightarrow (e = (s\text{-type mode}, s\text{-type mode}) \wedge (\tilde{\mathbf{x}}_j \text{ must be re-computed})). \end{aligned} \quad (50)$$

Proof 8. The proof is similar to the proof of proposition (6)

In summary, the computation of the maximal solution for (32) is obtained by running the following hybrid automaton

$$\mathcal{H} = (\mathcal{Q}, \mathcal{E}, \mathcal{S}, \mathcal{J}, \mathcal{F}, \mathcal{D}, \mathcal{P}) \quad (51)$$

where:

1. $\mathcal{Q} = \mathcal{Q}_m \cup \mathcal{Q}_s$ is a finite set of modes. $\mathcal{Q}_m$ gathers m -type modes, active over time intervals $[t_j, t_{j+1}] \in \overline{\mathcal{I}_m}$. Denote in this case the active continuous state vector by $\bar{x}_{(m)}$, obtained using (40). $\mathcal{Q}_s$ gathers s -type modes active over time intervals $[t_j, t_{j+1}] \in \overline{\mathcal{I}_s}$. Denote the active continuous state vector by $\bar{x}_{(s)}$, obtained using (47).
2. $\mathcal{E} \subseteq \mathcal{Q} \times \mathcal{Q}$ is the set of the transitions. According to propositions 5–8, the elements of $\mathcal{E}$ are of type $e_{ms} = (m\text{-type}, s\text{-type})$, $e_{sm} = (s\text{-type}, m\text{-type})$ and $e_{ss} = (s\text{-type}, s\text{-type})$.
3. $\mathcal{S} = \{t'_e, e \in \mathcal{E}\}$ is the collection of switching time instants. According to (36), and proposition 4 we have $\mathcal{S} = \{\text{Inf}(\mathbf{I}_m), \mathbf{I}_m \in \underline{\mathcal{I}_M}\} \cup \{\text{Inf}(\mathbf{I}_s), \mathbf{I}_s \in \overline{\mathcal{I}_S}\}$.
4. $\mathcal{J} = \{J_e, e = (q, q') \in \mathcal{E}\}$ is the collection of reset functions. Since we are dealing with the maximal solution of (32), and since the latter is cooperative, the discontinuous state jump is given, for all transitions, by $\bar{x}_{(q')}(t'_e{}^+) = \text{Sup}(\bar{x}_{(q)}(t'_e{}^-))$.
5. $\mathcal{F} = \{\bar{f}_{(q)}, q \in \mathcal{Q}\}$ is the collection of the field vectors.
6. $\mathcal{D}$ is the state space of (1).
7. $\mathcal{P} = \mathbf{p} = [\underline{p}, \bar{p}]$ represents the uncertainty domain for model parameters for (1).

Finally, we can build the **Hybrid-Upper-Bounding** algorithm, given below, for computing component-wise upper bounds for the reachable set of (1) by running automaton (51). It relies on algorithm **Integrate-one-step-ahead**, which computes the one-step ahead solution for an uncertain differential equation.

Algorithm **Hybrid-Upper-Bounding** finds the initial discrete mode q at line 2 (algorithm **Initialize**). In the *while* loop, it integrates the ODE ($\bar{f}_{(q)}$) until a transition occurs. For all modes, numerical integration is done using ITM.

Transition detection is performed at line 6 by algorithm **Switch-Cond**, given below. It implements mode switching according to propositions 6–8. If this is the case (boolean *-transition-* is *true*), algorithm **Switch-Cond** also returns the new discrete state q' and instantiate parameter vectors π_i .

Algorithm Hybrid-Upper-Bounding

Input: $t_0, t_{n_T}, \mathcal{F}, \bar{x}_0, \mathbf{p}$

Output: $\bar{x}_1, \dots, \bar{x}_{n_T}, \tilde{x}_0, \dots, \tilde{x}_{n_T-1}$

1. $j := 0$
2. $q := \text{Initialize}(f, \bar{x}_0, \mathbf{p})$
3. *while* ($j < n_T$) *do*
4. $\{\Delta t_j, \bar{x}_{j+1}, \tilde{x}_j\} := \text{Integrate-one-step-ahead}(q, \mathcal{F}, \bar{x}_j, \{\pi_i, i = 1, \dots, n\})$
5. $\{-switching-, q', \{\pi_i, i = 1, \dots, n\}\} := \text{Switch-Cond}(q, \tilde{x}_j, \{g\}, \mathbf{p})$
6. *if* (*-switching-*) *then*
7. *if* (q' of *m*-type) *then*
8. $q := q'; j := j + 1$
9. *else* (q' of *s*-type)
10. $q := q'$
11. *endif*
12. *else*
13. $j := j + 1$
14. *endif*
15. *end*

Algorithm Switch-Cond

input: $q, \tilde{\mathbf{x}}_j, \{g\}, \mathbf{p}$

output: $\text{-switching-}, q', \{\pi_i, i = 1, \dots, n\}$

1. for i, k do
2. Store partial derivatives signs
3. if $\left(\exists i^* \exists k^* \mid 0 \in \mathbf{g}_{i^*, k^*}(\tilde{\mathbf{x}}_j, \mathbf{p}, [t_i, t_{i+1}])\right)$ then
4. $\mathcal{I}^* \leftarrow i^*, \mathcal{K}^*(i^*) \leftarrow k^*$
5. endif
6. end do
7. if $(\mathcal{I}^* \neq \emptyset)$ then
8. $q' \leftarrow$ Read mode number $(\mathcal{I}^*, \mathcal{K}^*(.))$ % q' of s -type
9. if $(q' \neq q)$, then
10. $\text{-switching-} := \text{true}$
11. $\pi_i := \overline{\Delta}^{i^*}(\mathbf{p}), (i^* \in \mathcal{I}^*), \{i = 1, \dots, n\}$
12. endif
13. else % $\mathcal{I}^* = \emptyset$
14. $q' \leftarrow$ Read mode number % q' of m -type
15. if $(q' \neq q)$, then
16. $\text{-switching-} := \text{true}$
17. $\pi_i := \overline{\Delta}^i(p), \{i = 1, \dots, n\}$
18. endif

5. Applications

5.1. Reachability of uncertain nonlinear monotone system using comparison principles

This example is taken from molecular system biology. Consider a non-linear dynamical model, which describes Mitogen-Activated Protein Kinase cascades [45]

$$\begin{cases} \dot{x}_1 = & -(v_2x_1)/(k_2 + x_1) + v_0gx_5 + v_1 \\ \dot{x}_2 = & (v_6(y_{tot} - x_2 - x_3))/(k_6 + (y_{tot} - x_2 - x_3)) - (v_3x_1x_2)/(k_3 + x_2) \\ \dot{x}_3 = & (v_4x_1(y_{tot} - x_2 - x_3))/(k_4 + (y_{tot} - x_2 - x_3)) - (v_5x_3)/(k_5 + x_3) \\ \dot{x}_4 = & (v_{10}(z_{tot} - x_4 - x_5))/(k_{10} + (z_{tot} - x_4 - x_5)) - (v_7x_3x_4)/(k_7 + x_4) \\ \dot{x}_5 = & (v_8x_3(z_{tot} - x_4 - x_5))/(k_8 + (z_{tot} - x_4 - x_5)) - (v_9x_5)/(k_9 + x_5) \end{cases} \quad (52)$$

When ITM is used directly on system (52), the size of the enclosures blows up rapidly if a very small uncertainty (0.01%) is considered for parameter vector or if relative uncertainty on initial state vector is larger than 3%, as shown in Fig. 1.

We will now show how to compute the reachable set using theorem 2. The incidence graph obtained as indicated in section 2 is depicted in Fig. 2 ; it exhibits no negative cycles. According to proposition 1, system (52) is monotone with respect to the orthant S_ϵ where $\epsilon = (1, -1, 1, -1, 1)$. Using a change of coordinate $z = \text{Diag}(\epsilon)x$, it can be converted to a cooperative system, and theorem 2 applies.

Let us consider uncertainty on parameter and initial state vectors: $x_1 = [20, 170]$, $x_2 = [0, 100]$, $x_3 = [1000, 1200]$, $x_4 = [0, 10]$, $x_5 = [250, 300]$, $v_0 = [0.0015, 0.0016]$, $v_1 = [0.09, 0.1]$, $v_2 = [1.2, 1.3]$, $v_3 = v_4 = [0.064, 0.065]$, $v_5 = v_6 = v_9 = v_{10} = [5, 5.1]$, $v_7 = v_8 = [0.06, 0.07]$, $y_{tot} = [1200, 1201]$, $z_{tot} = [300, 301]$, $k_2 = [200, 201]$, $k_3 = k_4 = k_5 = k_6 = [1200, 1201]$, $k_7 = k_8 = k_9 = k_{10} = [300, 301]$. Interval Hermite Obreschkoff method with variable time step as implemented in the open source VNODE software [46] is used for solving the initial value problems for the bracketing systems obtained using the comparison principles detailed in section 3. Fig. 3 plots the time history of the x_2 component of the reachable set as obtained in both cases where parametric uncertainty is taken or not taken into account. It is clear that our bounding approach successfully computes an over approximation for the reachable set. Furthermore, it is easy to see that since parameter components appear only once in the algebraic expressions of (52), the bounding systems are feasible. Then, the enclosures obtained using our method are tight (up to the accuracy of the guaranteed numerical integration method used).

5.2. Reachability of uncertain nonlinear monotone system using our hybrid bounding approach

Consider the thermal model of a material sample submitted to a multi-harmonic signal [47]. State vector $x \in \mathbb{R}^{13}$ stands for temperature and t denotes time. The state equation is as follows:

$$\begin{cases} \dot{x}_1 &= \alpha_1(x_2 - 2x_1 + u_0 + \sum_{l=1 \dots 5} u_l \sin(2^{l-1}\omega_0 t + \varphi_0)) \\ \dot{x}_2 &= 2\alpha_1(x_1 - (1 + \frac{\rho_1}{\rho_2})x_2 + \frac{\rho_1}{\rho_2}x_3) \\ \dot{x}_3 &= 2(p_0 + p_1x_3)(x_4 - x_3 + p_2\frac{\delta_2}{\rho_2}(x_2 - x_3)) \\ \dot{x}_i &= (p_0 + p_1x_i)(x_{i+1} - 2x_i + x_{i-1}) \quad i = 4, \dots, 9 \\ \dot{x}_{10} &= 2(p_0 + p_1x_{10})(x_9 - x_{10} + p_2\frac{\delta_2}{\rho_2}(x_{11} - x_{10})) \\ \dot{x}_{11} &= 2\alpha_2(x_{12} - (1 + \frac{\rho_3}{\rho_2})x_{11} + \frac{\rho_3}{\rho_2}x_{10}) \\ \dot{x}_{12} &= \alpha_3(x_{13} - 2x_{12} + x_{11}) \\ \dot{x}_{13} &= 2\alpha_3(x_{12} - (1 + \frac{\rho_3}{\rho_4})x_{13} + \frac{\rho_3}{\rho_4}u_0) \end{cases} \quad (53)$$

Parameter vector $p = [p_0 \ p_1 \ p_2]^T$ is taken in the set $\mathbf{p} = [0.7, 1.23]\text{s}^{-1} \times [0.01, 0.015]\text{s}^{-1}\text{K}^{-1} \times [0.23, 0.64]\text{mW}^{-1}\text{K}$ and initial state vector domain is taken as $\mathbf{x}_{0_i} = [90, 110]^\circ\text{C}$.

The other parameters are assumed known; they take the following values: $\alpha_1 = 5.44\text{s}^{-1}$, $\alpha_2 = \alpha_3 = 5.7\text{s}^{-1}$, $u_0 = 100^\circ\text{C}$, $u_1 = 10.14^\circ\text{C}$, $u_2 = 13.84^\circ\text{C}$, $u_3 = 17.4^\circ\text{C}$, $u_4 = 23.87^\circ\text{C}$, $u_5 = 30.81^\circ\text{C}$, $\omega_0 = 0.00314\text{Hz}$, $\varphi_1 = 1.225$, $\varphi_2 = 1.103$, $\varphi_3 = 0.836$, $\varphi_4 = 0.470$, $\varphi_5 = 0.269$, $\rho_1/\rho_2 = 0.1777$, $\rho_3/\rho_4 = 0.0003$, $\rho_3/\rho_2 = 0.0910$, $\delta_2/\rho_2 = 2.2857\text{Wm}^{-1}\text{K}^{-1}$.

When one uses interval Hermite-Obreschkoff series with variable step control as implemented in the VNODE software [28] directly on the uncertain system (53), the computed enclosures blow up after one or two time steps (in fact VNODE software stops), as long as parameter vector p is taken uncertain, even with very small uncertainty.

Since system (53) is cooperative, we can use the hybrid bracketing technique introduced in section 4. To build the automaton (51) characterizing the *upper* bounding systems for (53), we need to study the signs of the partial derivatives $(\partial f_i / \partial p_k)$. Note that parameters p_0 , p_1 and p_2 appear in f_3 and f_{10} , and parameters p_0 and p_1 appear in f_i , $i = 4 \dots 9$. In addition, the signs of the partial derivatives $(\partial f_i / \partial p_0)$ and $(\partial f_i / \partial p_1)$ are similar. Hence, we have to monitor the sign of only 10 partial derivatives: $(\partial f_i / \partial p_0)$, $i = 3, \dots, 10$, and $(\partial f_{i'} / \partial p_2)$, $i' = 3$ and $i' = 10$. Now, according to the sign of the latter, i.e. using functions $\Delta^{i^*}(\cdot)$ defined in rule 2, the upper bounding systems are obtained by replacing parameter components in each algebraic expression of $f_{i'}$ either by their upper or by lower bound, for *m*-type modes, or by using whole parameter uncertainty domain, for *s*-type modes. Since there are 10 partial derivatives to monitor and 3 possible values for the parameter components (lower bound / upper bound / whole uncertainty interval), the set $\mathcal{Q}$ of discrete modes contains 3^{10} elements, and we merely use a word of ternary digits of length 10, to number the modes. Note however, that not all of them may be activated.

Fig. 4 shows the switching sequence for the hybrid automaton which derives the upper component-wise bounds of the reachable set of (53), as generated by algorithm **Hybrid-Upper-Bounding**. Some modes are active on very short time intervals. Fig. 5 magnifies the switching sequence around $t = 60\text{s}$. In fact, such modes are *s*-type modes which are usually active only over one or two integration time intervals.

The automaton which derives the lower component-wise bounds is obtained in a similar manner. The switching sequence for this automaton is shown on Fig. 6.

Note that both initial state vector and parameter vector are taken uncertain with large uncertainties. Fig. 7 shows the time history of the x_{12} component of the reachable set. Obviously, even for very large parameter boxes the hybrid bracketing method successfully computes the reachable set.

Since parameter p_k appears in several f_i , the enclosures obtained for the reachable set are not tight, as already discussed in section 3.3. This fact is emphasized in Fig. 7, where one can see typical state trajectories obtained with extreme values for p and starting from $\underline{x}$ or $\bar{x}$. It is clear that the enclosures derived by our hybrid bounding method are not reached by any of these trial trajectories. Note however, that the overestimation is not induced by our hybrid bounding approach, but by rules 1 and 2 used for deriving the bracketing systems for each mode.

6. Conclusion

In this paper we have addressed the issue of computing the reachable set for uncertain nonlinear continuous *monotone* dynamical systems. We have shown that for this type of systems, the hybrid bounding approach developed in our previous work can be improved in the sense that upper and lower component-wise bounds can be computed separately. A pleasant consequence is that mode switching needs not use whole solution sets and is thus more efficient. Along the lines of our previous work, we have re-formulated our hybrid bounding approach and also proposed an improved rule for obtaining the bracketing systems. Our new method handles successfully nonlinear, non autonomous, monotone systems even in presence of large uncertainty in both initial state and parameter vectors.

The rule used for obtaining the bounding systems works component-wise, therefore, these systems are generally not feasible. This leads to over-approximations in the computed reachable sets. This issue should be addressed to improve the performance of our method. Moreover, it should be interesting to study the stability of the size of the enclosures derived with the hybrid bounding method.

Here, we restricted our hybrid bounding approach to systems which are monotone with respect to some orthant of $\mathbb{R}^n$. Future work should address the case of monotone systems with respect to cones which are not limited to orthants of $\mathbb{R}^n$.

Finally, our hybrid bounding method should be extended to truly hybrid systems, those with monotone continuous dynamics.

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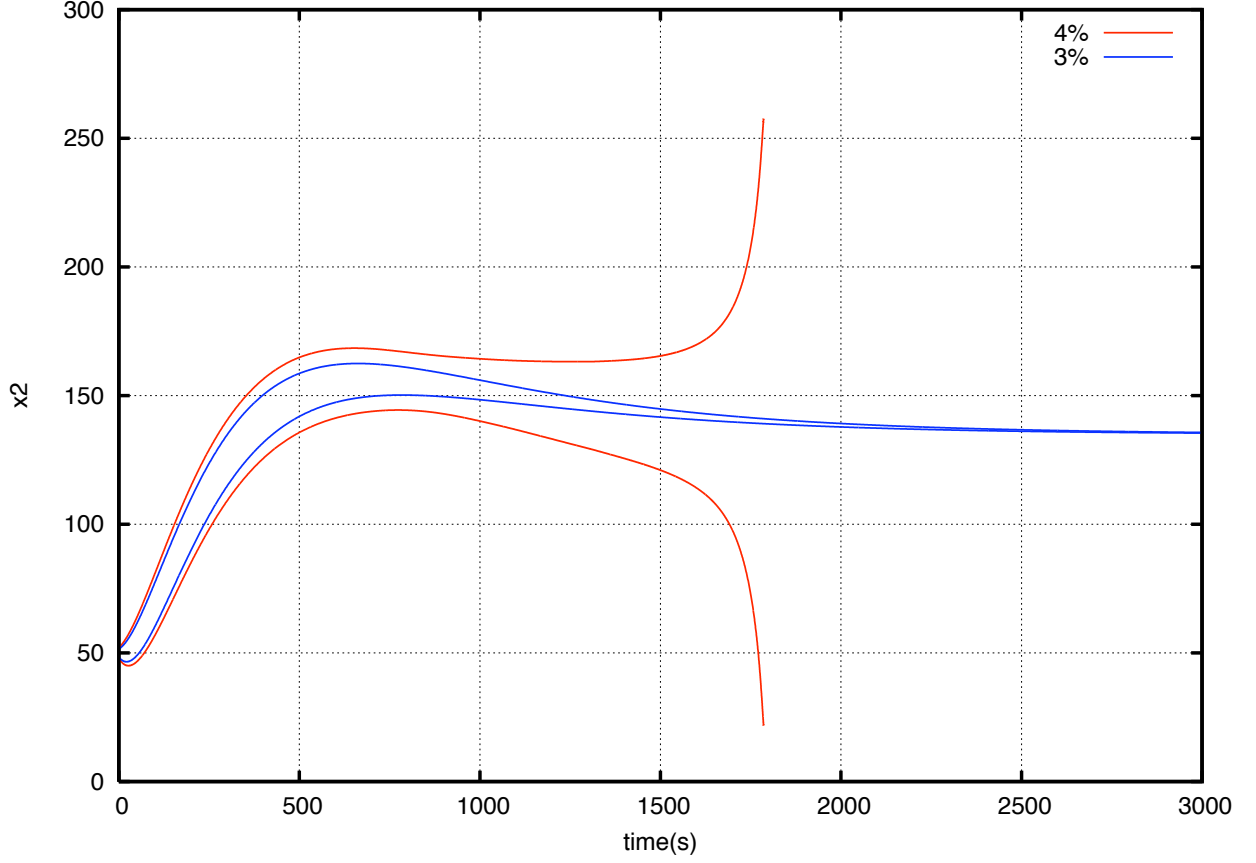


Figure 1: Time history of the x_2 component of the reachable set of (52) with no parametric uncertainty. The curve labelled 3% corresponds to 3% uncertainty on initial state vector, whereas the one labelled 4% corresponds to 4% uncertainty on initial state vector. Both curves are obtained with ITM (3%: CPU time = 22.44s PIV 2GHz). Without parametric uncertainty, ITM diverges as soon as the size of the domain for initial state vector is larger than 3%.

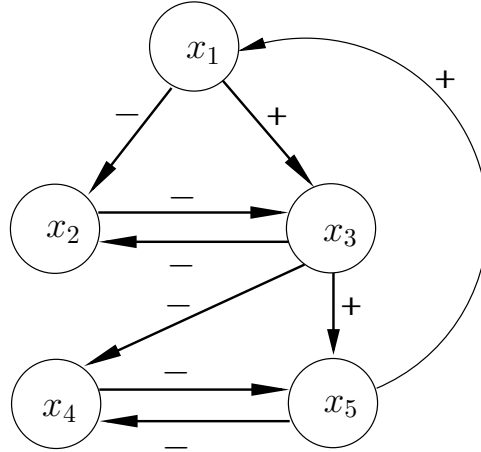


Figure 2: Incidence graph for system (52). It exhibits no negative cycles. System (52) is monotone with respect to the orthant $S_\epsilon = \{\epsilon_i x_i \geq 0\}$, $\epsilon = (1, -1, 1, -1, 1)$.

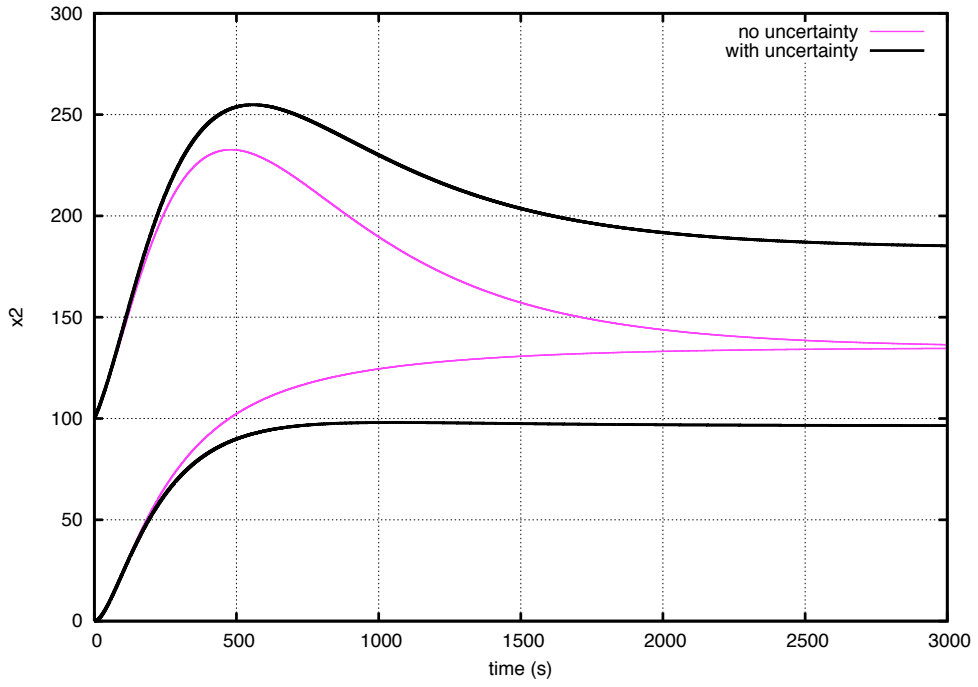


Figure 3: Time history of the x_2 component of the reachable set of (52) as obtained with theorem 2, with an initial domain for state vector of size 100%. The curve labelled 'no uncertainty' corresponds to no uncertainty in the parameter vector (CPU time = 38.26s PIV 2GHz) and the one labelled 'with uncertainty' corresponds to the presence of uncertainty in the parameter vector (CPU time = 38.58s PIV 2GHz).

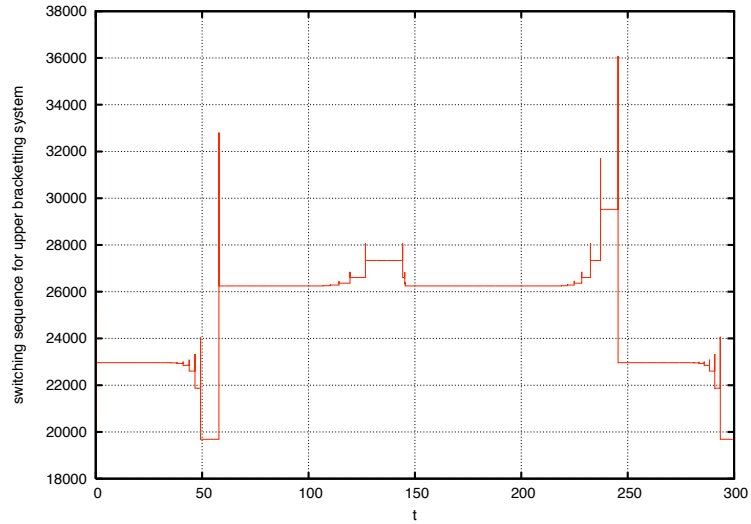


Figure 4: Switching sequence for the hybrid automaton which drives the upper bounding system for (53).

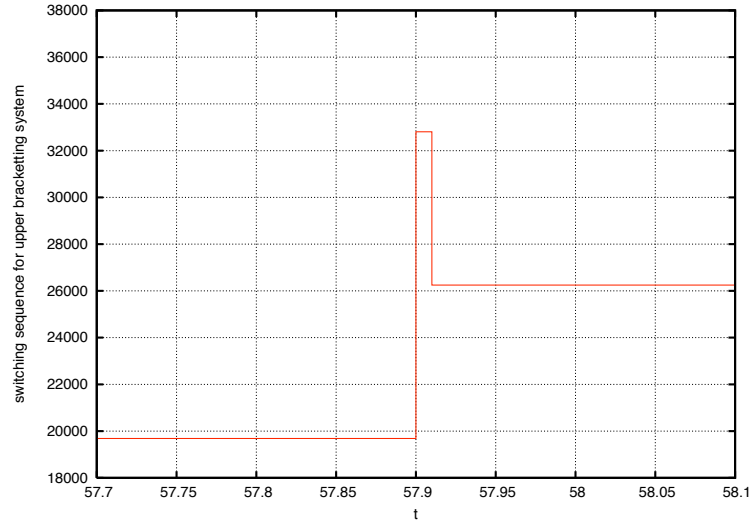


Figure 5: Zoom on the switching sequence for the hybrid automaton which drives the upper bounding system for (53), around $t = 60$ s.

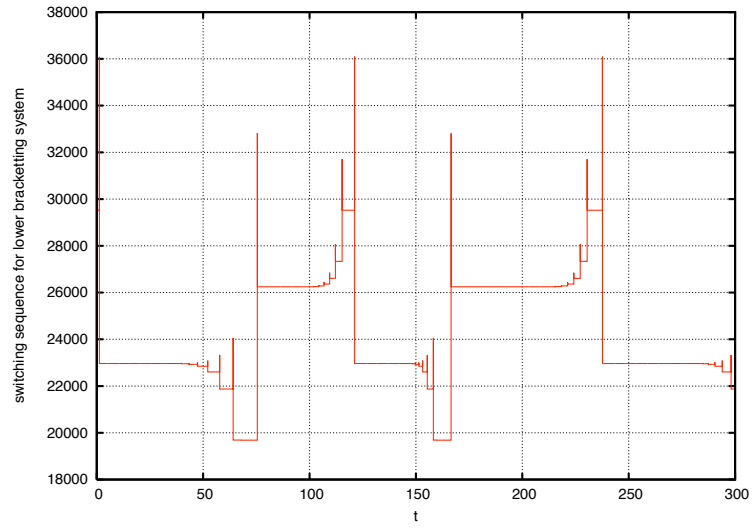


Figure 6: Switching sequence for the hybrid automaton which drives the lower bounding system for (53).

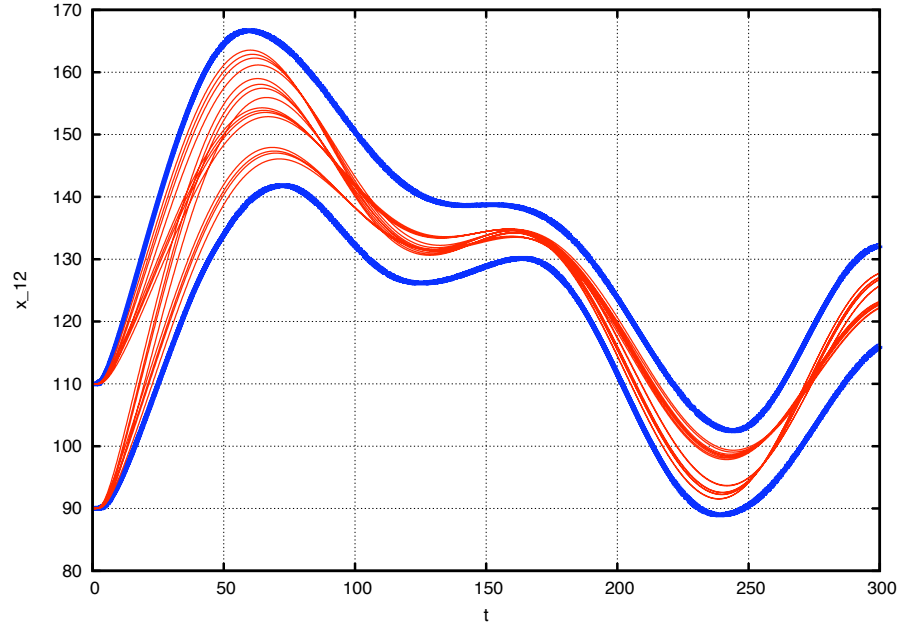


Figure 7: Reachable set of (53) with $X_{0_i} = [90, 110]^\circ\text{C}$ and $P_0 = [0.73, 1.23]\text{s}^{-1} \times [0.23, 0.64]\text{mW}^{-1}\text{K}$, as given by the hybrid bounding approach. Time history of x_{12} component of the reachable set (thick lines). (CPU time 67.8s PIV 2Ghz). See also trial trajectories obtained with extreme values for parameter and initial state vectors (thin lines).